

Daanish Uddin Khan

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15/08/2026

Education

Lahore University of Management Sciences (LUMS) — Lahore, Punjab

Jun 2025

Bachelor of Science: Computer Science

GPA: 3.26

Relevant Coursework: Introduction to AI, Data Structures, Network-Centric Computing, Machine Learning, Operating Systems, Algorithms, Principles and Techniques of Data Science, Software Engineering, Internet of Things, Cloud Development, Advanced Topics in Machine Learning

Publications

JURY: A Framework for Evaluating Chemical Language Models (Under Review)

Jan 2026 – Jul 2026

Journal of Cheminformatics

- Designed JURY, a four-probe evaluation framework spanning training-free to fully-supervised prediction, to separate what a frozen chemical language model encodes from what a trained prediction head recovers.
- Benchmarked ten pretrained chemical language models across three transformer architectures plus an ECFP fingerprint baseline on 73 ADMET datasets in DeepPK.
- Found that no single model dominates across probes and that ECFP outperforms all ten models on regression tasks, showing models differ in the form they hold chemistry rather than the amount.

Interpreting Molecular Transformers with Sparse Autoencoders (Under Review)

Jan 2026 – Aug 2026

Artificial Intelligence Chemistry

- Developed InterpretMolT, a method pairing sparse autoencoders with counterfactual single-atom edits to causally test whether a feature’s structural association is one it actually depends on.
- Applied framework to six pretrained molecular transformers finding that 38–46% of statistically significant feature-structure associations failed causal necessity tests.
- Identified a dichotomy in which a sparse minority of features performs over 93% of representational reconstruction while showing the weakest causal dependence on chemical structure.

Sustainability Analysis of LLM Web Search Engines (Accepted) ([Paper](#)) ([GitHub](#))

Jun 2025 – Oct 2025

The Web Conference 2026

- Built automated measurement pipeline to trace ChatGPT and Claude’s retrieval behavior across 1,000 queries, deploying controlled test website with instrumented logging to audit caching patterns and multi-page navigation.
- Discovered both engines function as meta-search wrappers, Claude had 92% overlap with Brave and GPT heavily favored Google results.
- Developed human-equivalent cost model showing LLM engines per-query page footprint far exceeds the 2–4 page human baseline, with 10–15 minute cache lifetimes.

Bridging IoT Gaps in Developing Regions with LLMs (Accepted) ([Paper](#)) ([GitHub](#))

Feb 2024 – May 2025

IEEE DCOSS-IoT 2026

- Built Kissan-Dost, a multilingual WhatsApp chatbot that converts live IoT sensor readings and weather into plain-language agricultural guidance via modular RAG+LLM pipeline with proactive alerts in Urdu, Punjabi, and Sindhi.
- Engineered ESP-NOW sensor network transmitting real-time field data to ESP32 gateway with cloud orchestration; integrated weather forecasts, market trends, and agricultural policies for context-rich insights.
- Led 90-day, two-site pilot deployment on farms near Lahore across three phases (baseline, dashboard, chatbot); evaluated system on 99 sensor-grounded queries with >90% correctness and sub-second latency.

Research Experience

German Research Center for Artificial Intelligence (DFKI) — Remote

Jul 2025 – Present

Research Assistant, Collaborating with RPTU Kaiserslautern

- Led evaluation-methodology design and benchmarking for pretrained chemical language models.
- Causally tested molecular transformer features using sparse autoencoders paired with counterfactual editing.
- Developing a unified contrastive framework evaluating five molecular representation modalities on 200M molecules.
- Engineered Mixture-of-Experts pipeline to automate molecular modality selection for downstream prediction.

SysNet Lab, LUMS

Feb 2024 – Oct 2025

Research Assistant

- Co-led a 90-day, two-site field deployment of an agricultural IoT-LLM chatbot near Lahore.
- Conducted interviews and thematic analysis across three study phases to evaluate trust and adoption.
- Built an automated measurement pipeline to trace LLM search-engine retrieval behavior across 1,000 queries.
- Compared LLM-fetched URLs against major search engine indices to assess reliance on existing infrastructure.

CSALT Lab, LUMS

Sep 2023 – May 2024

Research Intern

- Contributed to a novel Urdu question-answering dataset by translating SQuAD2.0 via the EATS methodology.
- Evaluated translation quality, acknowledged as a contributor in a publication at LREC-COLING 2024.
- Assessed multilingual LLM fact-checking for misinformation detection using Google's Fact Check Explorer.
- Executed Llama 2 inference runs across datasets to analyze model reliability for underserved regions.

Industry Experience

Systems Limited — Lahore, Punjab

Jun 2024 – Aug 2024

Intern — Cloud App Development

- Contributed to achieving project milestones by supporting the Cloud App Development team.
- Assisted in developing front-end features using React, contributing to a user-friendly interface.
- Participated in daily stand-up meetings, gaining practical insights into agile development processes.

Teaching Experience

Teaching Assistant — CS-677 Internet of Things

Jan 2025 – May 2025

- Part of a 2-member TA team supporting a graduate-level IoT course.
- Designed, conducted, and graded 4 hands-on IoT labs to familiarize students with core IoT hardware concepts.
- Oversaw students through semester-long IoT projects and provided consistent technical and academic support.

Teaching Assistant — CS-360 Software Engineering

Jan 2025 – May 2025

- Guided the semester-long projects of 4 student groups, with weekly check-ins covering the full SDLC.
- Worked within a team of 10 TAs to support 233 students, graded exams, and evaluated assignments.
- Provided mentorship and academic support through dedicated office hours and online channels.

Teaching Assistant — CS-535 Machine Learning

Sep 2024 – Dec 2024

- Collaborated with 10 teaching assistants for a graduate-level Machine Learning course with 135 students.
- Evaluated quizzes and coding assignments while offering guidance through office hours and online platforms.
- Supported students during a half-semester project and ensured adherence to academic policies.

Skills & Interests

- **Programming:** C++ (Advanced), C (Advanced), Python (Advanced), JavaScript/TypeScript (Advanced).
- **Cloud & IoT:** AWS (Lambda, DynamoDB, EC2, S3), Google Cloud. ESP32/ESP-NOW sensor networks.
- **AI/ML:** PyTorch, TensorFlow, Keras. LLMs, RAG, UDA. Pruning, quantization, knowledge distillation.
- **Research:** Literature review, benchmarking, experimental design. Analysis w/ Pandas, NumPy, Matplotlib.
- **Web Development:** MERN stack, Bootstrap. Puppeteer for automated testing.
- **Technical Communication:** Research paper writing for top-tier conferences and technical documentation.
- **Languages:** English (Fluent), Urdu (Fluent)

References

- **Dr. Muhammad Hamad Alizai**
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- **Dr. Agha Ali Raza**
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